



AQUATIC CONNECTIVITY

Tidal Stream Crossing Survey

DATA FORM

DATABASE ENTRY BY _____

ENTRY DATE _____

DATA ENTRY REVIEWED BY _____

REVIEW DATE _____

CROSSING DATA

Crossing Code _____ Local ID (Optional) _____

Date Observed (00/00/0000) _____ Start Time _____ End Time _____ Lead Observer _____

Town/County _____ Stream/River _____

Road _____ Type ☐ MULTILANE ☐ PAVED ☐ UNPAVED ☐ DRIVEWAY ☐ TRAIL ☐ RAILROADGPS Coordinates (Decimal degrees) °N Latitude — °W Longitude

Location Description

Crossing Type	Number of Culverts/Bridge Cells
☐ BRIDGE ☐ CULVERT ☐ MULTIPLE CULVERT ☐ FORD ☐ NO CROSSING ☐ REMOVED CROSSING	
☐ BURIED STREAM ☐ INACCESSIBLE ☐ PARTIALLY INACCESSIBLE ☐ NO UPSTREAM CHANNEL ☐ BRIDGE ADEQUATE	

Tide Stage ☐ LOW SLACK TIDE ☐ LOW EBB TIDE ☐ LOW FLOOD TIDE ☐ UNKNOWN ☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Tide Prediction Time of nearest low tide _____ Tide chart _____

Flow Conditions ☐ DEWATERED ☐ UNUSUALLY LOW ☐ TYPICAL LOW FLOW ☐ MODERATE FLOW ☐ HIGH FLOWStream Type ☐ SALT MARSH CREEK ☐ SALT/BRACKISH FLOW-THROUGH STREAM ☐ FRESHWATER TIDAL

Salinity (Optional) _____ ppt

Crossing Condition ☐ NEW ☐ OK ☐ POOR ☐ FAILING ☐ UNKNOWNVisible Utilities ☐ NONE ☐ OVERHEAD WIRES ☐ WATER/SEWER PIPE ☐ GAS LINE ☐ OTHER (DESCRIBE IN COMMENTS SECTION)Alignment ☐ FLOW ALIGNED ☐ SKEWED (>45°) ☐ Road Fill Height (ft.) (Top of culvert to road surface; bridge = 0) _____ Road Flooded at High Tide ☐ YES ☐ NO

Downstream Channel Width (ft.) _____ Pool Width (ft.) _____ Tidal Range (ft.) _____

Upstream Channel Width (ft.) _____ Pool Width (ft.) _____ Tidal Range (ft.) _____

Vegetation Above/Below ☐ COMPARABLE ☐ SLIGHTLY DIFFERENT ☐ MODERATELY DIFFERENT ☐ VERY DIFFERENT ☐ UNKNOWN

Photo File #s Outlet _____ Downstream _____ Inlet _____ Upstream _____

Other _____

Repeat Survey ☐ Yes ☐ No, New Survey ☐ Unknown Past Restoration Project? ☐ Completed ☐ Proposed ☐ UnknownOwner Type ☐ Federal ☐ State ☐ Local Govt ☐ Public Utility ☐ Private ☐ Tribe ☐ Other ☐ Unknown

Crossing Notes/Comments _____

STRUCTURE 1

Tide Gate Type ☐ NO TIDE GATE ☐ STOP LOGS ☐ FLAP GATE ☐ SLUICE GATE ☐ SELF-REGULATING
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Tide Gate Barrier Severity ☐ NO TIDE GATE ☐ MINOR ☐ MODERATE ☐ SEVERE ☐ NO AQUATIC PASSAGE

Outlet Materials (Select all options that apply) ☐ CONCRETE ☐ STONE ☐ WOOD ☐ METAL (SMOOTH) ☐ METAL (CORRUGATED)
☐ PLASTIC (SMOOTH) ☐ PLASTIC (CORRUGATED) ☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Outlet Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN ☐ CLOGGED/COLLAPSED/SUBMERGED

Outlet Dimensions (ft.) A. Width _____ B. Height _____ C. Substrate/Water Width _____ D. Water Depth _____

E. Abutment Height (Type 7 bridges only) _____ High Tide Water Depth _____ Spring Tide Water Depth _____

Outlet Perch (ft.) Low Tide _____ High Tide _____ **Outlet Armoring** ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Inlet Type ☐ PROJECTING ☐ MITERED TO SLOPE ☐ FLUSH (NOT MITERED) ☐ HEADWALL ☐ WING WALL(S) ☐ HEADWALL & WING WALL(S)
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Inlet Materials (Select all options that apply) ☐ CONCRETE ☐ STONE ☐ WOOD ☐ METAL (SMOOTH) ☐ METAL (CORRUGATED)
☐ PLASTIC (SMOOTH) ☐ PLASTIC (CORRUGATED) ☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Inlet Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN ☐ CLOGGED/COLLAPSED/SUBMERGED

Inlet Dimensions (ft.) A. Width _____ B. Height _____ C. Substrate/Water Width _____ D. Water Depth _____

E. Abutment Height (Type 7 bridges only) _____ High Tide Water Depth _____ Spring Tide Water Depth _____

Inlet Perch (ft.) Low Tide _____ High Tide _____ **Inlet Armoring** ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Structure Length (Overall length from inlet to outlet in ft.) _____

Relative Water Depth ☐ <0.10 ☐ 0.10-0.24 ☐ 0.25-0.49 ☐ 0.50-0.74 ☐ 0.75-0.99 ☐ ≥1.0

Structure Substrate Type (Pick one) ☐ NONE ☐ MUCK/SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ UNKNOWN

Structure Substrate Matches Stream ☐ NONE ☐ COMPARABLE ☐ CONTRASTING ☐ NOT APPROPRIATE (e.g. RIP RAP) ☐ UNKNOWN

Structure Substrate Coverage ☐ NONE ☐ 25%-50% ☐ 50%-75% ☐ 75%-99% ☐ 100% ☐ UNKNOWN

Structure Slope (Relative to Channel) ☐ COMPARABLE ☐ SUBSTANTIALLY FLATTER ☐ SUBSTANTIALLY STEEPER

Other Barrier Type ☐ NONE ☐ SEDIMENT BLOCKAGE ☐ DEBRIS ☐ FENCING ☐ PIPES ☐ DEFORMATION ☐ FREE FALL
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Other Barrier Severity ☐ NONE ☐ MINOR ☐ MODERATE ☐ SEVERE ☐ NO AQUATIC PASSAGE

Dry Passage for Terrestrial Wildlife ☐ YES ☐ NO ☐ UNKNOWN **Height above Dry Passage (ft.)** _____

Structure Notes/Comments _____

ADDITIONAL CONDITIONS

082319

OUTLET

Outlet Perch (ft.) Low Tide _____ High Tide _____ Outlet Armoring ■ NONE ■ NOT EXTENSIVE ■ EXTENSIVE

INLET

Inlet Perch (ft.) Low Tide _____ High Tide _____ Inlet Armoring ■ NONE ■ NOT EXTENSIVE ■ EXTENSIVE

ADDITIONAL CONDITIONS

Dry Passage for Terrestrial Wildlife	YES	NO	UNKNOWN	Height above Dry Passage (ft.)
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Structure Notes/Comments

STRUCTURE 3

Tide Gate Type ☐ NO TIDE GATE ☐ STOP LOGS ☐ FLAP GATE ☐ SLUICE GATE ☐ SELF-REGULATING
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Tide Gate Barrier Severity ☐ NO TIDE GATE ☐ MINOR ☐ MODERATE ☐ SEVERE ☐ NO AQUATIC PASSAGE

OUTLET

Outlet Materials (Select all options that apply) ☐ CONCRETE ☐ STONE ☐ WOOD ☐ METAL (SMOOTH) ☐ METAL (CORRUGATED)
☐ PLASTIC (SMOOTH) ☐ PLASTIC (CORRUGATED) ☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Outlet Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN ☐ CLOGGED/COLLAPSED/SUBMERGED

Outlet Dimensions (ft.) A. Width _____ B. Height _____ C. Substrate/Water Width _____ D. Water Depth _____
 E. Abutment Height (Type 7 bridges only) _____ High Tide Water Depth _____ Spring Tide Water Depth _____

Outlet Perch (ft.) Low Tide _____ High Tide _____ **Outlet Armoring** ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

INLET

Inlet Type ☐ PROJECTING ☐ MITERED TO SLOPE ☐ FLUSH (NOT MITERED) ☐ HEADWALL ☐ WING WALL(S) ☐ HEADWALL & WING WALL(S)
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Inlet Materials (Select all options that apply) ☐ CONCRETE ☐ STONE ☐ WOOD ☐ METAL (SMOOTH) ☐ METAL (CORRUGATED)
☐ PLASTIC (SMOOTH) ☐ PLASTIC (CORRUGATED) ☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Inlet Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN ☐ CLOGGED/COLLAPSED/SUBMERGED

Inlet Dimensions (ft.) A. Width _____ B. Height _____ C. Substrate/Water Width _____ D. Water Depth _____
 E. Abutment Height (Type 7 bridges only) _____ High Tide Water Depth _____ Spring Tide Water Depth _____

Inlet Perch (ft.) Low Tide _____ High Tide _____ **Inlet Armoring** ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

ADDITIONAL CONDITIONS

Structure Length (Overall length from inlet to outlet in ft.) _____

Relative Water Depth ☐ <0.10 ☐ 0.10-0.24 ☐ 0.25-0.49 ☐ 0.50-0.74 ☐ 0.75-0.99 ☐ ≥1.0

Structure Substrate Type (Pick one) ☐ NONE ☐ MUCK/SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ UNKNOWN

Structure Substrate Matches Stream ☐ NONE ☐ COMPARABLE ☐ CONTRASTING ☐ NOT APPROPRIATE (e.g. RIP RAP) ☐ UNKNOWN

Structure Substrate Coverage ☐ NONE ☐ 25%-50% ☐ 50%-75% ☐ 75%-99% ☐ 100% ☐ UNKNOWN

Structure Slope (Relative to Channel) ☐ COMPARABLE ☐ SUBSTANTIALLY FLATTER ☐ SUBSTANTIALLY STEEPER

Other Barrier Type ☐ NONE ☐ SEDIMENT BLOCKAGE ☐ DEBRIS ☐ FENCING ☐ PIPES ☐ DEFORMATION ☐ FREE FALL
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Other Barrier Severity ☐ NONE ☐ MINOR ☐ MODERATE ☐ SEVERE ☐ NO AQUATIC PASSAGE

Dry Passage for Terrestrial Wildlife ☐ YES ☐ NO ☐ UNKNOWN **Height above Dry Passage (ft.)** _____

Structure Notes/Comments _____

STRUCTURE 4

Tide Gate Type ☐ NO TIDE GATE ☐ STOP LOGS ☐ FLAP GATE ☐ SLUICE GATE ☐ SELF-REGULATING
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Tide Gate Barrier Severity ☐ NO TIDE GATE ☐ MINOR ☐ MODERATE ☐ SEVERE ☐ NO AQUATIC PASSAGE

Outlet Materials (Select all options that apply) ☐ CONCRETE ☐ STONE ☐ WOOD ☐ METAL (SMOOTH) ☐ METAL (CORRUGATED)
☐ PLASTIC (SMOOTH) ☐ PLASTIC (CORRUGATED) ☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Outlet Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN ☐ CLOGGED/COLLAPSED/SUBMERGED

Outlet Dimensions (ft.) A. Width _____ B. Height _____ C. Substrate/Water Width _____ D. Water Depth _____
 E. Abutment Height (Type 7 bridges only) _____ High Tide Water Depth _____ Spring Tide Water Depth _____

Outlet Perch (ft.) Low Tide _____ High Tide _____ **Outlet Armoring** ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Inlet Type ☐ PROJECTING ☐ MITERED TO SLOPE ☐ FLUSH (NOT MITERED) ☐ HEADWALL ☐ WING WALL(S) ☐ HEADWALL & WING WALL(S)
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Inlet Materials (Select all options that apply) ☐ CONCRETE ☐ STONE ☐ WOOD ☐ METAL (SMOOTH) ☐ METAL (CORRUGATED)
☐ PLASTIC (SMOOTH) ☐ PLASTIC (CORRUGATED) ☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Inlet Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN ☐ CLOGGED/COLLAPSED/SUBMERGED

Inlet Dimensions (ft.) A. Width _____ B. Height _____ C. Substrate/Water Width _____ D. Water Depth _____
 E. Abutment Height (Type 7 bridges only) _____ High Tide Water Depth _____ Spring Tide Water Depth _____

Inlet Perch (ft.) Low Tide _____ High Tide _____ **Inlet Armoring** ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

Structure Length (Overall length from inlet to outlet in ft.) _____

Relative Water Depth ☐ <0.10 ☐ 0.10-0.24 ☐ 0.25-0.49 ☐ 0.50-0.74 ☐ 0.75-0.99 ☐ ≥1.0

Structure Substrate Type (Pick one) ☐ NONE ☐ MUCK/SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ UNKNOWN

Structure Substrate Matches Stream ☐ NONE ☐ COMPARABLE ☐ CONTRASTING ☐ NOT APPROPRIATE (e.g. RIP RAP) ☐ UNKNOWN

Structure Substrate Coverage ☐ NONE ☐ 25%-50% ☐ 50%-75% ☐ 75%-99% ☐ 100% ☐ UNKNOWN

Structure Slope (Relative to Channel) ☐ COMPARABLE ☐ SUBSTANTIALLY FLATTER ☐ SUBSTANTIALLY STEEPER

Other Barrier Type ☐ NONE ☐ SEDIMENT BLOCKAGE ☐ DEBRIS ☐ FENCING ☐ PIPES ☐ DEFORMATION ☐ FREE FALL
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Other Barrier Severity ☐ NONE ☐ MINOR ☐ MODERATE ☐ SEVERE ☐ NO AQUATIC PASSAGE

Dry Passage for Terrestrial Wildlife ☐ YES ☐ NO ☐ UNKNOWN **Height above Dry Passage (ft.)** _____

Structure Notes/Comments _____

ADDITIONAL CONDITIONS

OUTLET

Tide Gate Barrier Severity ■ NO TIDE GATE ■ MINOR ■ MODERATE ■ SEVERE ■ NO AQUATIC PASSAGE

Outlet Materials (Select all options that apply) ☐ CONCRETE ☐ STONE ☐ WOOD ☐ METAL (SMOOTH) ☐ METAL (CORRUGATED)
☐ PLASTIC (SMOOTH) ☐ PLASTIC (CORRUGATED) ☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Outlet Shape 1 2 3 4 5 6 7 FORD REMOVED UNKNOWN CLOGGED/COLLAPSED/SUBMERGED

Outlet Dimensions (ft.) A. Width _____ B. Height _____ C. Substrate/Water Width _____ D. Water Depth _____

E. Abutment Height (Type 7 bridges only) _____ **High Tide Water Depth** _____ **Spring Tide Water Depth** _____

Outlet Perch (ft.) Low Tide _____ High Tide _____ Outlet Armoring ■ NONE ■ NOT EXTENSIVE ■ EXTENSIVE

INLET

Inlet Type ☒ PROJECTING ☐ MITERED TO SLOPE ☐ FLUSH (NOT MITERED) ☐ HEADWALL ☐ WING WALL(S) ☐ HEADWALL & WING WALL(S)
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Inlet Materials (Select all options that apply) ☐ CONCRETE ☐ STONE ☐ WOOD ☐ METAL (SMOOTH) ☐ METAL (CORRUGATED)
☐ PLASTIC (SMOOTH) ☐ PLASTIC (CORRUGATED) ☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Inlet Shape ■ 1 ■ 2 ■ 3 ■ 4 ■ 5 ■ 6 ■ 7 ■ FORD ■ REMOVED ■ UNKNOWN ■ CLOGGED/COLLAPSED/SUBMERGED

Inlet Dimensions (ft.) A. Width _____ B. Height _____ C. Substrate/Water Width _____ D. Water Depth _____

E. Abutment Height (Type 7 bridges only) _____ **High Tide Water Depth** _____ **Spring Tide Water Depth** _____

Inlet Perch (ft.) Low Tide _____ High Tide _____ Inlet Armoring ■ NONE ■ NOT EXTENSIVE ■ EXTENSIVE

ADDITIONAL CONDITIONS

Structure Length (Overall length from inlet to outlet in ft.)

Relative Water Depth ■ <0.10 ■ 0.10-0.24 ■ 0.25-0.49 ■ 0.50-0.74 ■ 0.75-0.99 ■ ≥1.0

Structure Substrate Type (Pick one) ☐ NONE ☐ MUCK/SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ UNKNOWN

Structure Substrate Matches Stream ■ NONE ■ COMPARABLE ■ CONTRASTING ■ NOT APPROPRIATE (e.g. RIP RAP) ■ UNKNOWN

Structure Substrate Coverage ■ NONE ■ 25%-50% ■ 50%-75% ■ 75%-99% ■ 100% ■ UNKNOWN

Structure Slope (Relative to Channel) ■ COMPARABLE ■ SUBSTANTIALLY FLATTER ■ SUBSTANTIALLY STEEPER

Other Barrier Type ☐ NONE ☐ SEDIMENT BLOCKAGE ☐ DEBRIS ☐ FENCING ☐ PIPES ☐ DEFORMATION ☐ FREE FALL
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Other Barrier Severity ■ NONE ■ MINOR ■ MODERATE ■ SEVERE ■ NO AQUATIC PASSAGE

Dry Passage for Terrestrial Wildlife	YES	NO	UNKNOWN	Height above Dry Passage (ft.)
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Structure Notes/Comments

STRUCTURE 6

Tide Gate Type ☐ NO TIDE GATE ☐ STOP LOGS ☐ FLAP GATE ☐ SLUICE GATE ☐ SELF-REGULATING
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Tide Gate Barrier Severity ☐ NO TIDE GATE ☐ MINOR ☐ MODERATE ☐ SEVERE ☐ NO AQUATIC PASSAGE

OUTLET

Outlet Materials (Select all options that apply) ☐ CONCRETE ☐ STONE ☐ WOOD ☐ METAL (SMOOTH) ☐ METAL (CORRUGATED)
☐ PLASTIC (SMOOTH) ☐ PLASTIC (CORRUGATED) ☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Outlet Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN ☐ CLOGGED/COLLAPSED/SUBMERGED

Outlet Dimensions (ft.) A. Width _____ B. Height _____ C. Substrate/Water Width _____ D. Water Depth _____

E. Abutment Height (Type 7 bridges only) _____ High Tide Water Depth _____ Spring Tide Water Depth _____

Outlet Perch (ft.) Low Tide _____ High Tide _____ **Outlet Armoring** ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

INLET

Inlet Type ☐ PROJECTING ☐ MITERED TO SLOPE ☐ FLUSH (NOT MITERED) ☐ HEADWALL ☐ WING WALL(S) ☐ HEADWALL & WING WALL(S)
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Inlet Materials (Select all options that apply) ☐ CONCRETE ☐ STONE ☐ WOOD ☐ METAL (SMOOTH) ☐ METAL (CORRUGATED)
☐ PLASTIC (SMOOTH) ☐ PLASTIC (CORRUGATED) ☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Inlet Shape ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ FORD ☐ REMOVED ☐ UNKNOWN ☐ CLOGGED/COLLAPSED/SUBMERGED

Inlet Dimensions (ft.) A. Width _____ B. Height _____ C. Substrate/Water Width _____ D. Water Depth _____

E. Abutment Height (Type 7 bridges only) _____ High Tide Water Depth _____ Spring Tide Water Depth _____

Inlet Perch (ft.) Low Tide _____ High Tide _____ **Inlet Armoring** ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

ADDITIONAL CONDITIONS

Structure Length (Overall length from inlet to outlet in ft.) _____

Relative Water Depth ☐ <0.10 ☐ 0.10-0.24 ☐ 0.25-0.49 ☐ 0.50-0.74 ☐ 0.75-0.99 ☐ ≥1.0

Structure Substrate Type (Pick one) ☐ NONE ☐ MUCK/SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ UNKNOWN

Structure Substrate Matches Stream ☐ NONE ☐ COMPARABLE ☐ CONTRASTING ☐ NOT APPROPRIATE (e.g. RIP RAP) ☐ UNKNOWN

Structure Substrate Coverage ☐ NONE ☐ 25%-50% ☐ 50%-75% ☐ 75%-99% ☐ 100% ☐ UNKNOWN

Structure Slope (Relative to Channel) ☐ COMPARABLE ☐ SUBSTANTIALLY FLATTER ☐ SUBSTANTIALLY STEEPER

Other Barrier Type ☐ NONE ☐ SEDIMENT BLOCKAGE ☐ DEBRIS ☐ FENCING ☐ PIPES ☐ DEFORMATION ☐ FREE FALL
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Other Barrier Severity ☐ NONE ☐ MINOR ☐ MODERATE ☐ SEVERE ☐ NO AQUATIC PASSAGE

Dry Passage for Terrestrial Wildlife ☐ YES ☐ NO ☐ UNKNOWN **Height above Dry Passage (ft.)** _____

Structure Notes/Comments _____

STRUCTURE 7

Tide Gate Type

- NO TIDE GATE
- STOP LOGS
- FLAP GATE
- SLUICE GATE
- SELF-REGULATING

☐ NO TIDE GATE ☐ STOP LOSS ☐ TEMP GATE ☐ SECTOR GATE ☐ SLEEP REGULATING

☐ OTHER (DESCRIBE IN COMMENTS SECTION)

OUTLET

Tide Gate Barrier Severity ■ NO TIDE GATE ■ MINOR ■ MODERATE ■ SEVERE ■ NO AQUATIC PASSAGE

Outlet Materials (Select all options that apply) ☐ CONCRETE ☐ STONE ☐ WOOD ☐ METAL (SMOOTH) ☐ METAL (CORRUGATED)
☐ PLASTIC (SMOOTH) ☐ PLASTIC (CORRUGATED) ☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Outlet Shape 1 2 3 4 5 6 7 FORD REMOVED UNKNOWN CLOGGED/COLLAPSED/SUBMERGED

Outlet Dimensions (ft.) A. Width _____ B. Height _____ C. Substrate/Water Width _____ D. Water Depth _____
E. Abutment Height (Type 7 bridges only) _____ High Tide Water Depth _____ Spring Tide Water Depth _____

Outlet Perch (ft.) Low Tide _____ High Tide _____ Outlet Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

INLET

Inlet Type ☐ PROJECTING ☐ MITERED TO SLOPE ☐ FLUSH (NOT MITERED) ☐ HEADWALL ☐ WING WALL(S) ☐ HEADWALL & WING WALL(S)
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Inlet Materials (Select all options that apply) ☐ CONCRETE ☐ STONE ☐ WOOD ☐ METAL (SMOOTH) ☐ METAL (CORRUGATED)
☐ PLASTIC (SMOOTH) ☐ PLASTIC (CORRUGATED) ☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Inlet Shape ■ 1 ■ 2 ■ 3 ■ 4 ■ 5 ■ 6 ■ 7 ■ FORD ■ REMOVED ■ UNKNOWN ■ CLOGGED/COLLAPSED/SUBMERGED

Inlet Dimensions (ft.) **A. Width** _____ **B. Height** _____ **C. Substrate/Water Width** _____ **D. Water Depth** _____
E. Abutment Height (Type 7 bridges only) _____ **High Tide Water Depth** _____ **Spring Tide Water Depth** _____

Inlet Perch (ft.) Low Tide _____ High Tide _____ Inlet Armoring ☐ NONE ☐ NOT EXTENSIVE ☐ EXTENSIVE

ADDITIONAL CONDITIONS

Structure Length (Overall length from inlet to outlet in ft.) _____

Relative Water Depth <0.10 0.10-0.24 0.25-0.49 0.50-0.74 0.75-0.99 ≥1.0

Structure Substrate Type (Pick one) ☐ NONE ☐ MUCK/SILT ☐ SAND ☐ GRAVEL ☐ COBBLE ☐ BOULDER ☐ BEDROCK ☐ UNKNOWN

Structure Substrate Matches Stream ■ NONE ■ COMPARABLE ■ CONTRASTING ■ NOT APPROPRIATE (e.g. RIP RAP) ■ UNKNOWN

Structure Substrate Coverage ■ NONE ■ 25%-50% ■ 50%-75% ■ 75%-99% ■ 100% ■ UNKNOWN

Structure Slope (Relative to Channel) ■ COMPARABLE ■ SUBSTANTIALLY FLATTER ■ SUBSTANTIALLY STEEPER

Other Barrier Type ☐ NONE ☐ SEDIMENT BLOCKAGE ☐ DEBRIS ☐ FENCING ☐ PIPES ☐ DEFORMATION ☐ FREE FALL
☐ OTHER (DESCRIBE IN COMMENTS SECTION)

Other Barrier Severity ■ NONE ■ MINOR ■ MODERATE ■ SEVERE ■ NO AQUATIC PASSAGE

Dry Passage for Terrestrial Wildlife ☐ YES ☐ NO ☐ UNKNOWN **Height above Dry Passage (ft.)** _____

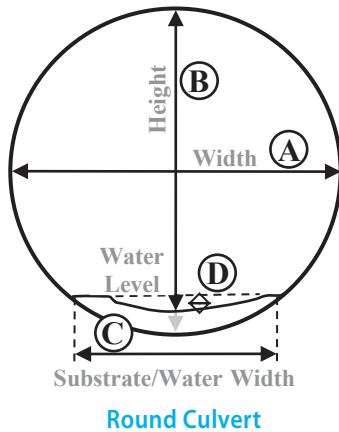
Structure Notes/Comments

Structure Shape & Dimensions

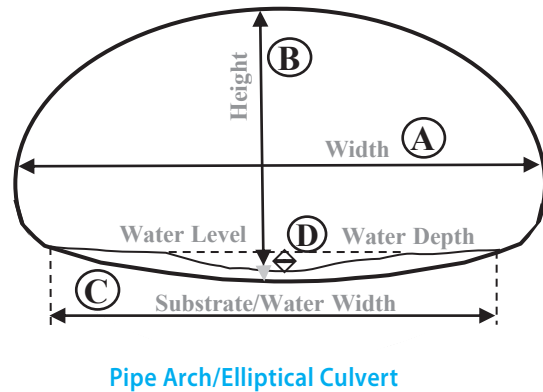
- 1) Select the Structure Shape number from the diagrams below and record it on the form for Inlet and Outlet Shape.
- 2) Record on the form in the appropriate blanks dimensions **A**, **B**, **C** and **D** as shown in the diagrams;
C captures the width of water or substrate, whichever is wider; for dry culverts without substrate, C = 0.
D is the depth of water -- be sure to measure inside the structure; for dry culverts, D = 0.
- 3) Record Structure Length (**L**). (Record abutment height (**E**) only for Type 7 Structures.)
- 4) For multiple culverts, also record the Inlet and Outlet shape and dimensions for each additional culvert.

NOTE: Culverts 1, 2 & 4 may or may not have substrate in them, so height measurements (B) are taken from the level of the "stream bed", whether that bed is composed of substrate or just the inside bottom surface of a culvert (grey arrows below show measuring to bottom, black arrows show measuring to substrate).

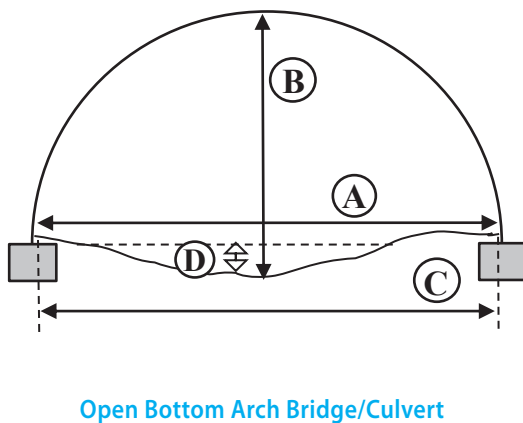
1



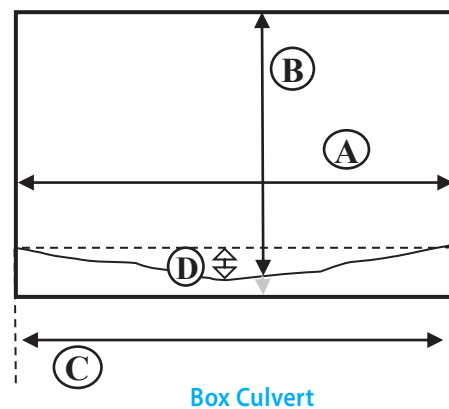
2



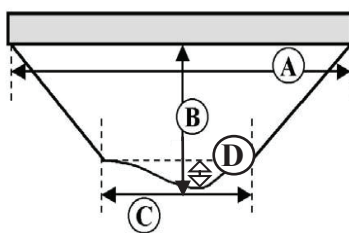
3



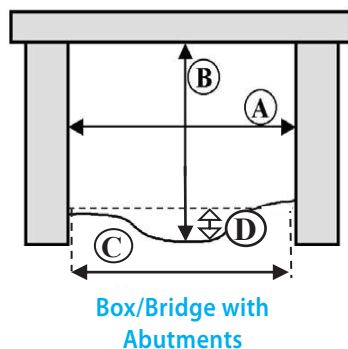
4



5



6



7

